

Dear Hiring Team,

I am writing to express my interest in graduate opportunities at Google. I am a final-year Theoretical and Mathematical Physics student at Trinity College Dublin, with a strong interest in programming and in building well-designed, scalable solutions to real problems.

Across both my academic work and industry experience, I have repeatedly taken on unfamiliar and technically demanding problems, requiring me to quickly develop new skills and approaches. This has ranged from building Python-based automation technologies in research environments to designing and implementing backend systems in a fast-moving startup. I am particularly drawn to understanding how systems behave at a fundamental level and how thoughtful design choices lead to robust and maintainable solutions.

Outside of formal work, I spend a significant amount of time programming personal projects driven by curiosity. These include building modular Python systems such as a local voice-controlled assistant, experimenting with machine learning models applied to physical systems, and developing simulation tools for traffic flow and infrastructure modelling. I enjoy working through problems end to end, from system design and experimentation to testing, documentation, and iteration, and I often use these projects as a way to explore new ideas or tools in depth.

Much of my recent experience has involved working as part of small, cross-functional teams. At Grinds360, I contributed as a developer alongside engineers, educators, and leadership during the development and launch of key platform features. This required clear communication, shared ownership of problems, and adapting technical decisions to evolving requirements and constraints.

A significant part of my role at Grinds360 also involved close engagement with users. By speaking directly with teachers and understanding how students interacted with the platform, I was able to help translate feedback into concrete improvements to the user experience, including work on the main website and the exam creation system. I enjoy asking the right questions, identifying user needs, and using those insights to inform product and design decisions.

I would welcome the opportunity to bring my technical background, curiosity, and collaborative approach to Google, while continuing to grow as an engineer working on impactful products.

Thank you for your time and consideration.